

Cai, Yifan

EDUCATION

University of Pennsylvania

Master of Science in Computer & Information Science

Philadelphia, PA

Aug 2024

Advisor: Dr. Linh Thi Xuan Phan

Ph.D. Student in Computer & Information Science

Aug 2021 – Present

Advisor: Dr. Linh Thi Xuan Phan

Research Interest: Dependability of distributed systems and real-time cyber-physical systems

Shanghai Jiao Tong University

Shanghai, China

Bachelor of Engineering, Software Engineering, System Software Track

Jun 2020

Honors/Awards: SJTU Academic Excellency Scholarships (2017, 2018 and 2019)

EXPERIENCES

Roblox

San Mateo, CA

Research Intern, supervised by Dr. Andreas Haeberlen and Dr. Linh Thi Xuan Phan

May 2024 – Aug 2024

- Designed a partitioning system for large-scale multiplayer gaming environments, optimizing server resource allocation while maintaining seamless player experience.

Alibaba Cloud

Sunnyvale, CA

Research Intern, supervised by Dr. Mo Sha and Dr. Sheng Wang

May 2023 – Aug 2023

- Studied data confidentiality, integrity, and freshness in heterogeneous memory architectures, while providing a unified interface to the applications.

Alibaba Cloud

Hangzhou, China

Research Intern, supervised by Dr. Sheng Wang and Dr. Wenchao Zhou

Sept 2020 – May 2021

- Designed and implemented a cloud database providing data and execution integrity that can be verified by the users.

University of Pennsylvania

Philadelphia, PA

Research Intern, Advisor: Dr. Vincent Liu

Jul 2019 – Dec 2019

- Studied the impact of disaggregated data centers on the design of relational databases.

TEACHING

Teaching Assistant, University of Pennsylvania

- CIS 505 Software Systems, Fall 2021
- CIS 505 Software Systems, Fall 2022

PUBLICATIONS

Uncertainty-Aware Resource Allocation for Multi-Path Programs with In-Kernel Predictions

Abigail Eisenklam, Carlos A. Montenegro G., Xian Wang, **Yifan Cai**, Robert Gifford, Linh Thi Xuan Phan, and Ricardo G. Sanfelice.
To appear at *38th European Conference on Real-Time Systems (ECRTS)*, 2026

Generative Profiling for Soft Real-Time Systems and its Applications to Resource Allocation

Georgiy A. Bondar, Abigail Eisenklam, **Yifan Cai**, Robert Gifford, Tushar Sial, Linh Thi Xuan Phan, and Abhishek Halder.
To appear at *32nd IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*, 2026

GeoShield: Byzantine Fault Detection and Recovery for Geo-Distributed Real-Time Cyber-Physical Systems

Yifan Cai and Linh Thi Xuan Phan

arXiv:2511.15031 [cs.CR]

Rasco: Resource Allocation and Scheduling Co-design for DAG Applications on Multicore

Abigail Eisenklam, Robert Gifford, Georgiy A. Bondar, **Yifan Cai**, Tushar Sial, Linh Thi Xuan Phan, and Abhishek Halder.
In *ACM Transactions on Embedded Computing Systems (TECS)*,
Special issue of ACM SIGBED International Conference on Embedded Software (EMSOFT), 2025

RoboRebound: Multi-Robot System Defense with Bounded-Time Interaction

Neeraj Gandhi, **Yifan Cai**, Andreas Haeberlen, and Linh Thi Xuan Phan

In the *20th European Conference on Computer Systems (EuroSys)*, 2025

Analysis of Long-term Average Behaviors of Probabilistic Task Systems

Yifan Cai, Linh Thi Xuan Phan, and P.S. Thiagarajan.

In *International Conference on Real-Time Networks and Systems (RTNS)*, 2024

Object-oriented Unified Encrypted Memory Management for Heterogeneous Memory Architectures

Mo Sha*, **Yifan Cai***, Sheng Wang, Linh Thi Xuan Phan, Feifei Li, and Kian-Lee Tan (* Equal contribution)

In *International Conference on Management of Data (SIGMOD)*, 2024

VeriDB: An SGX-based Verifiable Database.

Wenchao Zhou, **Yifan Cai**, Yanqing Peng, Sheng Wang, Ke Ma, and Feifei Li

In *International Conference on Management of Data (SIGMOD)*, 2021

Understanding the Effect of Data Center Resource Disaggregation on Production DBMSs.

Qizhen Zhang, **Yifan Cai**, Xinyi Chen, Sebastian Angel, Ang Chen, Vincent Liu, and Boon Thau Loo.

In *46th International Conference on Very Large Data Bases (VLDB)*, 2020.

Intra-day Forecast of Ground Horizontal Irradiance Using Long Short-Term Memory Network (LSTM)

Xiuhong Chen, Xianglei Huang, **Yifan Cai**, Haomin Shen, and Jiayue Lu.

In *Journal of the Meteorological Society of Japan. Ser. II*, 2020, 98(5): 945-957.

Rethinking Data Management Systems for Disaggregated Data Centers.

Qizhen Zhang, **Yifan Cai**, Sebastian Angel, Ang Chen, Vincent Liu, and Boon Thau Loo.

In *Conference on Innovative Data Systems Research (CIDR)*, 2020.

Consensus-based Data Statistics in Distributed Network Systems.

Yifan Cai, Jianping He, Wenbin Yu, and Xinpeng Guan.

In the *57th IEEE Conference on Decision and Control (CDC)*, 2018